

Analysis on Manifolds

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1.1 Conservative Fields and Locally Conservative Fields

Definition 1.1

Locally Conservative

Let $F : \Omega \rightarrow \mathbb{R}^d$ be a vector field. We say that F is **locally conservative** (meshamer mekomit) in Ω if for every $k, \ell = 1, \dots, d$,

$$\partial_k F_\ell = \partial_\ell F_k \iff \frac{\partial F_\ell}{\partial k} = \frac{\partial F_k}{\partial \ell}$$

Definition 1.2

C^k -smooth

Domain $\Omega \subset \mathbb{R}^d$ is **C^k -smooth** if for all $x \in \partial\Omega$ there exists a neighborhood $x \in V$ and a C^k -diffeomorphism $\varphi : B_1(0) \rightarrow V$ such that $\varphi(B_1(0) \cap \{x_d > 0\}) = V \cap \Omega$ and we call φ a rectifying map.

Informal: Ω is smooth if for all $p \in \partial\Omega$ there exists a neighborhood $p \in V$ such that $V \cap \Omega$ looks like a half circle.

Definition 1.3

Simply Connected

X a topological space is **simply connected** if is path connected and any loop ($\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = \gamma(1)$) in X can be continuously contracted a point (i.e., is homotopic to a constant loop, with no “holes”).

Theorem 1.4

If Ω is simply connected and F is locally conservative in Ω then F is a conservative in Ω .

Proof: In exercise 3. □

Notation

$$\int_\gamma \vec{F} d\vec{\ell} = \int_\gamma F_1 dx_1 + F_2 dx_2 + \dots + F_\ell dx_\ell$$

Exercise 1.5

Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \rightarrow V$ a rectifying map with $\text{supp}(F) \subseteq V$ ($\varphi(t, s) = (x(t, s), y(t, s))$). If $J_p > 0$ then $\varphi(t, 0)$ is a parametrization of $\partial\Omega \cap V$ with positive orientation (since we never use reflection).

Definition 1.6

Partition of Unity

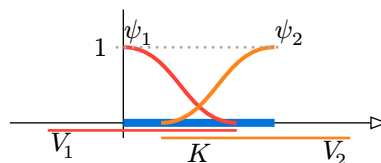
Let $K \subseteq \mathbb{R}^d$ be a compact set and $K \subset V_1 \cup V_2 \cup \dots \cup V_m$ a finite open cover. Then there exists C^k functions $\psi_1, \dots, \psi_m$ defined in a neighborhood of K such that

1. $\forall j, \psi_j \geq 0$
2. $\text{supp}(\psi_j) \subseteq V_j$
3. $\sum_{j=1}^m \psi_j = 1$ on K (meaning for all $x \in K$ we have $\sum_{j=1}^m \psi_j(x) = 1$)

Such collection $\{\psi_j\}_{j=1}^m$ is called **Partition of Unity** on K subordnatory to the cover $\{V_j\}_{j=1}^m$.

Example

Let $K = [0, 2]$, $V_1 = (-1, 1)$, $V_2 = (\frac{1}{2}, 3)$ so $K \subset V_1 \cup V_2$ and we define ψ_1, ψ_2



Notice that the above and Theorem 1.7 and it's proof were spread over lectures 6,7,8 and each definition / exercise was in the matching lecture for the part of the proof we used it.

Theorem 1.7

Green's Theorem

Let $\Omega \subset \mathbb{R}^2$ be a bounded smooth domain and Let $F = (P, Q)$ be a smooth vector field in a neighborhood of $\partial\Omega$ and $\partial\Omega$ be positively oriented (left leg rule) then

$$\oint_{\partial\Omega} P dx + Q dy = \oint_{\partial\Omega} \vec{F} d\vec{\ell} = \int_{\Omega} (\partial_x Q - \partial_y P) dx dy$$

Since $\gamma(t) = (x(t), y(t))$, $dx = x'(t) dt$, $dy = y'(t) dt$ we get

$$\int_{\gamma} P dx + Q dy = \int (P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)) dt = \int \left\langle \begin{pmatrix} P \\ Q \end{pmatrix}, \begin{pmatrix} x' \\ y' \end{pmatrix} \right\rangle dt$$

Proof:

1st case: Suppose $\text{supp}(F) = \overline{\{x \in \Omega \mid F(x) \neq 0\}} \subset \Omega$.

Since $F|_{\partial\Omega} = 0$ we attain that $\int_{\partial\Omega} \vec{F} d\vec{\ell} = 0$ so we need to show that the integral on the right is also zero.

Since Ω is a bounded smooth domain there exists $M \in \mathbb{R}_+$ so $\Omega \subset [-M, M] \times [-M, M]$ hence

$$\begin{aligned} \int_{\Omega} (\partial_x Q - \partial_y P) dx dy &= \int_{[-M, M] \times [-M, M]} (\partial_x Q - \partial_y P) dx dy \\ &= \int_{-M}^M \int_{-M}^M \partial_x Q dx dy - \int_{-M}^M \int_{-M}^M \partial_y P dy dx \\ &= \int_{-M}^M Q(M, y) - Q(-M, y) dy + \int_{-M}^M P(x, M) - P(x, -M) dx = 0 \end{aligned}$$

Where the second equal is Fubini's Theorem and the third one is Fundamental Theorem of Calculus.

2nd case: Assume $\Omega = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1, y > 0\}$ and $\text{supp}(F) \subset B_1(0)$.

First, we need to find a parametrization for our boundary: With $-1 \leq t \leq 1$ we have $\gamma(t) = (t, 0)$ with $\gamma'(t) = (1, 0)$ hence

$$\int_{\partial\Omega} P dx + Q dy = \int_{-1}^1 P(t, 0) \cdot 1 + Q(t, 0) \cdot 0 dt = \int_{-1}^1 P(t, 0) dt$$

We need to compute $\int (\partial_x Q - \partial_y P) dx dy$ over the upper half circle of and show that they are equal.

Over the upper half circle we can replace with rectangle (small enough) and use Fubini's Theorem and Fundamental Theorem of Calculus

$$\int \partial_x Q dx dy = \int_{(-1, 1) \times (0, 1)} \partial_x Q dx dy = \int_0^1 \int_{-1}^1 \partial_x Q dx dy = \int_0^1 \underbrace{Q(1, y)}_{=0} - \underbrace{Q(-1, y)}_{=0} dy = 0$$

Since we are out of our support. In the same way

$$-\int \partial_y P dx dy = -\int_{(-1, 1) \times (0, 1)} \partial_y P dx dy = -\int_{-1}^1 \int_0^1 \partial_y P dy dx = -\int_{-1}^1 \underbrace{P(x, 1)}_{=0} - P(x, 0) dx = \int_{-1}^1 P(x, 0) dx$$

Using a change of variable we finish this case.

End of Lecture 6 – 28/04

3rd case: Let $p \in \partial\Omega$, V a neighborhood of p and $\varphi : B_1(0) \rightarrow V$ a rectifying map with $\text{supp}(F) \subseteq V$ ($\varphi(t, s) = (x(t, s), y(t, s))$).

We can assume that $J_p > 0$ (if not, we can replace φ by reflection – simply flipping one coordinate).

Using Exercise 1.5 since $\text{supp}(F) \subseteq V$ we can write with the definition of line integral

$$\int_{\partial\Omega} P dx + Q dy = \int_{\partial\Omega \cap V} P dx + Q dy = \int_{-1}^1 P(\varphi(t, 0)) \frac{\partial x}{\partial t}(t, 0) + Q(\varphi(t, 0)) \frac{\partial y}{\partial t}(t, 0) dt$$

For the other side, we start by computing

$$J_{\varphi} = \det \begin{pmatrix} \frac{\partial x}{\partial t} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial t} & \frac{\partial y}{\partial s} \end{pmatrix} = \frac{\partial x}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial x}{\partial s} \frac{\partial y}{\partial t}$$

Since F is smooth we use change of variable to get

$$\begin{aligned}
\int_{\Omega} \partial_x Q - \partial_y P \, dx \, dy &= \int_{\Omega \cap V} \partial_x Q - \partial_y P \, dx \, dy \\
&= \int_{\mathbf{A}} (\partial_x Q \circ \varphi - \partial_y P \circ \varphi) J_{\varphi} \, dt \, ds \\
&= \int_{\mathbf{A}} (\partial_x Q \circ \varphi - \partial_y P \circ \varphi) \left(\frac{\partial x}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial x}{\partial s} \frac{\partial y}{\partial t} \right) dt \, ds \\
&= \int_{\mathbf{A}} \left((\partial_x Q \circ \varphi) \frac{\partial x}{\partial t} + (\partial_y Q \circ \varphi) \frac{\partial y}{\partial t} \right) \frac{\partial y}{\partial s} - \left((\partial_x Q \circ \varphi) \frac{\partial x}{\partial s} + (\partial_y Q \circ \varphi) \frac{\partial y}{\partial s} \right) \frac{\partial y}{\partial t} \\
&\quad + \left((\partial_x P \circ \varphi) \frac{\partial x}{\partial t} + (\partial_y P \circ \varphi) \frac{\partial y}{\partial t} \right) \frac{\partial x}{\partial s} - \left((\partial_x P \circ \varphi) \frac{\partial x}{\partial s} + (\partial_y P \circ \varphi) \frac{\partial y}{\partial s} \right) \frac{\partial x}{\partial t} \\
&\stackrel{(1)}{=} \int_{\mathbf{A}} \left[\frac{\partial(Q \circ \varphi)}{\partial t} \frac{\partial y}{\partial s} - \frac{\partial(Q \circ \varphi)}{\partial s} \frac{\partial y}{\partial t} + \frac{\partial(P \circ \varphi)}{\partial t} \frac{\partial x}{\partial s} - \frac{\partial(P \circ \varphi)}{\partial s} \frac{\partial x}{\partial t} \right] dt \, ds \\
&\stackrel{(2)}{=} \int_{\mathbf{A}} \left[\frac{\partial}{\partial t} \left((Q \circ \varphi) \frac{\partial y}{\partial s} \right) - \frac{\partial}{\partial s} \left((Q \circ \varphi) \frac{\partial y}{\partial t} \right) + \frac{\partial}{\partial t} \left((P \circ \varphi) \frac{\partial x}{\partial s} \right) - \frac{\partial}{\partial s} \left((P \circ \varphi) \frac{\partial x}{\partial t} \right) \right] dt \, ds \\
&= \int_{\mathbf{A}} \left[\frac{\partial}{\partial t} \left(\underbrace{(P \circ \varphi) \frac{\partial x}{\partial s} + (Q \circ \varphi) \frac{\partial y}{\partial s}}_{\tilde{Q}} \right) - \frac{\partial}{\partial s} \left(\underbrace{(P \circ \varphi) \frac{\partial x}{\partial t} + (Q \circ \varphi) \frac{\partial y}{\partial t}}_{\tilde{P}} \right) \right] dt \, ds
\end{aligned}$$

Where (1) is Chain Rule and (2) Leibniz Integral Rule (Notice the line break with the +).

If we take $\vec{G}(t, s) = (\tilde{P}, \tilde{Q})$ and since $\gamma(t) = (t, 0)$, $\dot{\gamma}(t) = (1, 0)$, the 2nd case gives us

$$\int_{\partial\Omega} \tilde{P} \, dt + \tilde{Q} \, ds = \int_{\mathbf{A}} \frac{\partial \tilde{Q}}{\partial t} - \frac{\partial \tilde{P}}{\partial s} \, dt \, ds \implies \int \tilde{P} \, dt + \tilde{Q} \, ds = \int_{-1}^1 \tilde{P} \, dt = \int_{-1}^1 \left[(P \circ \varphi) \frac{\partial y}{\partial t}(t, 0) + (Q \circ \varphi) \frac{\partial y}{\partial t}(t, 0) \right] dt$$

Which is indeed the left side.

In other words for the 3rd case: we pull our vector field back to the upper half circle and go back to the 2nd case.

End of Lecture 7 – 30/04

4rd case: For all $p \in \partial\Omega$ there exists a neighborhood V_p and a rectifying map $\varphi_p : B_1(0) \rightarrow V_p$.

Since $\partial\Omega \subset \bigcup_{p \in \partial\Omega} V_p$ and since $\partial\Omega$ is compact then there exists a finite open subcover $\partial\Omega \subset V_{p_1} \cup \dots \cup V_{p_m}$. Let $V_0 = \Omega$ and $\bar{\Omega} \subset V_0 \cup V_{p_1} \cup \dots \cup V_{p_m}$ and for simplicity of notations let us write $v_j = v_{\gamma_j}$.

There exists a partition of unity on $\bar{\Omega}$ subordinate to $\{V_j\}_{j=0}^m$.

Define $\vec{F}_j := \psi_j \cdot \vec{F}$ and $\text{supp}(\vec{F}_j) \subset V_j$ and we have

$$\sum_{j=1}^m \vec{F}_j = \sum_{j=1}^m (\psi_j \cdot \vec{F}) = \left(\sum_{j=1}^m \psi_j \right) \cdot \vec{F} \stackrel{\text{on } \bar{\Omega}}{=} 1 \cdot \vec{F} = \vec{F}$$

From the 1st case if we set $\vec{F}_j = (P_j, Q_j)$ we since $\text{supp}(\vec{F}_0) \subseteq \Omega$ that

$$\int_{\partial\Omega} \vec{F}_0 \, d\vec{\ell} = \int_{\gamma} \partial_X Q_0 - \partial_y P_0 \, dx \, dy$$

By the 3rd case we know that for each $1 \leq j \leq m$ we have

$$\int_{\partial\Omega} \vec{F}_j \, d\vec{\ell} = \int_{\gamma} \partial_x Q_j - \partial_y P_j \, dx \, dy$$

Summation gives us

$$\sum_{j=0}^m \int_{\partial\Omega} \vec{F}_j \, d\vec{\ell} = \sum_{j=0}^m \int_{\gamma} \partial_x Q_j - \partial_y P_j \, dx \, dy \implies \int_{\partial\Omega} \vec{F} \, d\vec{\ell} = \int_{\gamma} \partial_x Q - \partial_y P \, dx \, dy$$

□

Remark

Green Theorem works for other domains and for example in the exercise we will prove for rectangles.

Example

Suppose that F is locally conservative in $\Omega \subseteq \mathbb{R}^2$ so $F = (P, Q)$ and $\partial_y P = \partial_x Q$ from Green's theorem

$$\int_{\partial\Omega} \vec{F} \, d\vec{\ell} = 0$$